Debug the Cause

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach diagnosis before patching the result.

Model competing faults, keep specifications fixed and reserve the ticket-rule task for independent evidence.

Keep one habit: Look for a first divergence, a distinguishing prediction and retest evidence.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Compare **expected** and observed **trace**s to locate the first difference

Use debugging as **evidence**-based investigation of a stated procedure. Distinguish finding the first observed divergence from establishing a unique root cause.

The arithmetic examples show that two faults can fit one result and that a test may separate two explanations while still missing a different fault. Use paper **trace**s and preserve the original failing case throughout repair.

By the end, I can choose a test that separates two possible causes of an error.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- The points-machine **specification** and its observed **trace** on separate cards
- An **expected** and observed **trace** sheet with an instruction-version field
- Cards for add 4 and add original start, plus the ticket rule for independent assessment

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

Look for: **Trace** a loop by testing its condition before every repetition

Reference example, not the only acceptable wording: Tests: 2 true, 4 true, 6 true, 8 false. Three additions are three iterations. The loop reports 8; the fourth test does not run the body.

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

Look for: **Trace** an ordered algorithm using the current value after every instruction

Reference example, not the only acceptable wording: The **trace** is 3, 7, 14, 9. The doubling step uses 7, the current value after adding 4 to 3. It must not double the initial 3.

Accept a table, arrows or clearly labelled sentences that preserve the order.

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

Look for: Interpret an ordered sequence of sound-signal samples with a stated time interval

Reference example, not the only acceptable wording: Five samples, a 2 ms interval, and 8 ms from first to last. Consecutive pairs make gaps 0-2, 2-4, 4-6 and 6-8: four gaps between five samples.

The tests see 3 true, 5 true and 7 false. Two iterations occur; the body does not run after the false test. Report 7.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Repeating Efficiently

Algorithms as Recipes

Sound Becomes a Signal

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Starting at square 0, the target is square 3. Program: repeat twice {MOVE 1}; STOP. A teammate changes the target to 2 and calls the bug fixed. Repair the program and give two tests.

Concept to explain

Debugging finds the broken assumption or step before changing the solution.

Term	Meaning
bug	cause of incorrect or unexpected behavior
trace	record of steps and changing state
expected	result predicted before the test
actual	result observed during the test

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach debugging as a testable comparison

> Preserve the specification, expected trace and actual record before selecting a repair.

Begin with the points-machine divergence

A paper points machine must add 3 to its starting score, double the updated score, then subtract 2 and report. From 4 the **expected** values are 4, 7, 14 and 12. These instructions are the **specification**.

Calculate the **expected result** from them before looking at the machine **output**.

An observed run instead records 4, 7, 11 and 9. The final result is wrong, but changing the report from 9 to 12 would only hide this particular failure.

Compare the intermediate values to find where the behaviour first stopped matching the instructions.

The first difference is after the doubling instruction: **expected** 14, observed 11. The final subtraction is consistent with the observed 11, since 11 minus 2 is 9.

That later difference can be a consequence of the earlier fault rather than a second faulty subtraction. A **trace** narrows the investigation; it does not automatically reveal every cause.

Define a reproducible record and stable expectation

Record the **input**, the exact instruction version, **expected** values, observed values and first mismatch.

This makes the failure reproducible: someone else can follow the same procedure with the same starting value and check your **observation**.

Do not rewrite the **expected** column to match the **observed result**. If the **specification** itself seems ambiguous, clarify it first and record that change openly.

A test can judge an implementation only against a stated expectation.

Model the add-6 versus double comparison

A fictional paper calculator should double the current value, but a copied instruction behaves strangely.

After **input** 6, the faulty step reports 12. Hypothesis A: add 6. Hypothesis B: double.

Both hypotheses fit this first **observation**. A new **input** may be chosen.

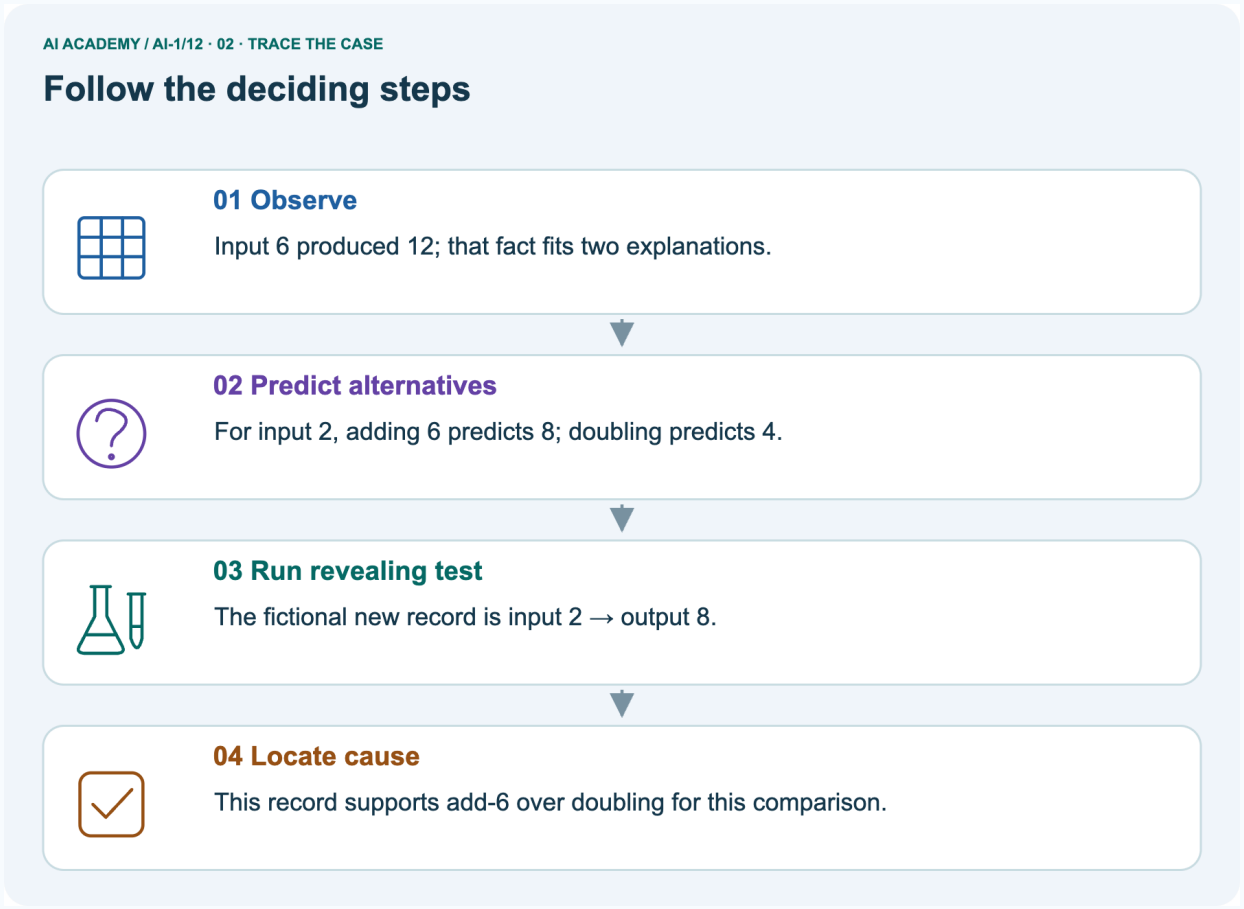
AI ACADEMY / AI-1/12

A revealing test separates explanations

	Input	Add 6 predicts	Double predicts
Original input	6	12	12
Recorded new input	2	8	4
Proposed input	0	6	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Repeating **input** 6 would not separate the two hypotheses because both again predict 12. **Input** 2 makes their **prediction**s different before the test is run.

The second record supports investigating an added constant, but inspecting the instruction is still needed to confirm the implementation.

Repeat an input that cannot distinguish the hypotheses

1. Change: Use **input** 6 again instead of 2.
2. **Prediction**: Both hypotheses predict 12; the test remains inconclusive.
3. Reason: A revealing test requires different predicted **output**s, not merely another recorded run.

Limit the diagnosis to the proposed alternatives

Two toy hypotheses do not exhaust every possible bug; a matching **output** alone cannot identify arbitrary hidden code.

Extend to competing faults in the full procedure

For the faulty doubling step, two explanations fit the observed change from 7 to 11. Perhaps the step adds 4. Perhaps it adds the original starting score, which was 4.

Repeating the same start 4 cannot separate those explanations because they make the same **prediction**.

Start score	Expected final result	If step adds 4 or original start
4	12	9 under either explanation
2	8	7 if add 4 or 5 if add original 2
5	14	10 if add 4 or 11 if add original 5

Starting at 2 produces 5 after the first addition. The intended double gives 10 and then reports 8. Adding 4 would give 9 and report 7. Adding the original 2 would give 7 and report 5.

The different **prediction**s make this a revealing test.

If the observed final result is 7, it supports add 4 over add original among these two explanations. Other faults could still give 7, so inspect the **actual** instruction or more checkpoints before claiming a unique cause.

Use the picture deliberately

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the **actual** procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

> **1. Input — information entering**
>
> **INPUT expected** path and **actual trace**

> **2. Process — the deciding step**
>
> **PROCESS** locate the first difference and test one cause

> **3. Output — the result**
>
> **OUTPUT** targeted correction plus repeated test

CORE IDEA Debugging finds the broken assumption or step before changing the solution.

UNDER THE HOOD Debugging is causal diagnosis. The first observable divergence narrows the faulty assumption, data, state, condition, or step.

Changing multiple things at once destroys **evidence** about which change repaired the behavior.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A paper robot hits a wall because one turn instruction assumes it faces north when it **actual**ly faces east.

STEP 1 — OBSERVE **expected** path and **actual trace**

STEP 2 — **TRACE** locate the first difference and test one cause

STEP 3 — CONCLUDE targeted correction plus repeated test

MODEL REASONING **Trace expected** and **actual** positions step by step. The first mismatch reveals the earliest plausible cause.

Change that instruction or assumption only, then rerun the identical maze to show whether the cause was correct.

DO NOT OVERCLAIM A successful new run does not prove the diagnosis if several things changed together.

What new **evidence** could change this answer?

Concrete teaching case

A robot should go from square one to square three. Its program is MOVE ONE, STOP.

Find the cause and repair one thing.

Exact explanation to model

It stops at square two because one move is missing. Add MOVE ONE before STOP and retest from the same start.

Explain the added worked example

Key terms

Term	Meaning
Specification	A statement of what a system is required to do
Expected result	The result worked out from the specification before running a test
Observed result	The result actually recorded during a test
First divergence	The earliest step where observed behaviour differs from expected behaviour
Regression test	A repeated check that previously correct behaviour still works after a change

Model the reasoning aloud

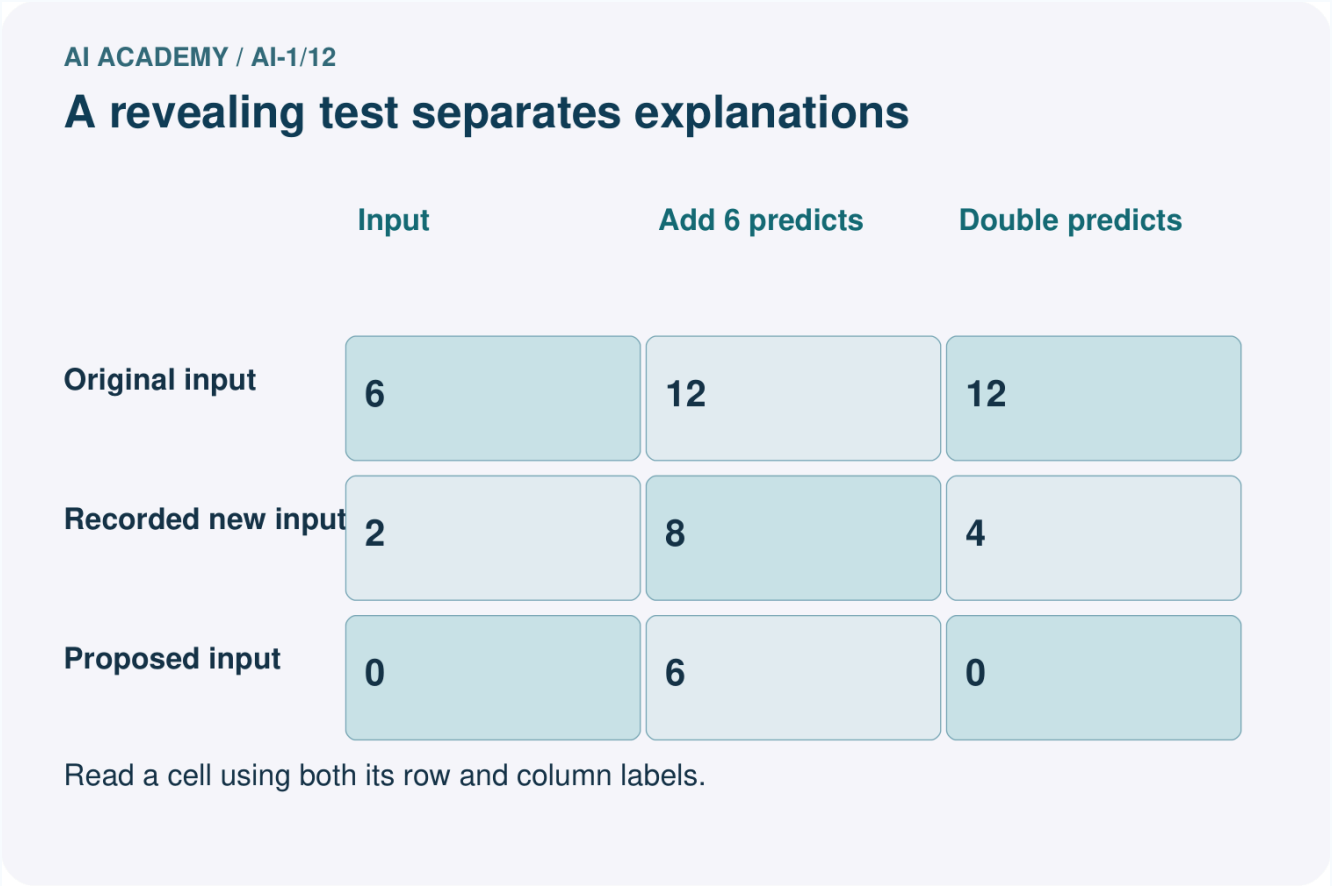
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Debug the Cause

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/12

Debug the Cause

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Observe

Input 6 produced 12; that fact fits two explanations.



2. Predict alternatives

For input 2, adding 6 predicts 8; doubling predicts 4.



3. Run revealing test

The fictional new record is input 2 -> output 8.



4. Locate cause

This record supports add-6 over doubling for this comparison.

CHANGE ONE THING

Use input 6 again instead of 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Observe	Input 6 produced 12; that fact fits two explanations.
2. Predict alternatives	For input 2, adding 6 predicts 8; doubling predicts 4.
3. Run revealing test	The fictional new record is input 2 → output 8.
4. Locate cause	This record supports add-6 over doubling for this comparison.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: In W1 the faulty middle step may add 3, or add the original starting value. Both explain 5 becoming 8 when the start is 3. Choose start 1 or start 3 for a revealing new test.

Predict the final result under each explanation and under the intended procedure.

Expected reasoning and fading help

Teacher check: Choose start 1. After add 2 the value is 3. Adding 3 then subtracting 1 reports 5; adding original 1 then subtracting 1 reports 3. Doubling then subtracting 1 should report 5.

This separates the two fault explanations, but a final 5 alone cannot separate add 3 from correct doubling on this **input**; retain the original failing start 3.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full **trace** only when the partial **trace** is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Debugging with revealing tests. Proposed explanation: “The last wrong number locates the fault”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Compare every aligned checkpoint; an early mismatch can cause several later wrong values.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Propose **input** 0 and give both predicted **output**s without inventing an **observed result**.

Add-6 predicts 6; doubling predicts 0. No **actual** result has been supplied for **input** 0.

Changed-case reasoning: Both hypotheses predict 12; the test remains inconclusive. A revealing test requires different predicted **output**s, not merely another recorded run.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A robot should go from square one to square three. Its program is MOVE ONE, STOP.

Write the question to explain: Find the cause and repair one thing.

Reveal after **prediction**s: It stops at square two because one move is missing. Add MOVE ONE before STOP and retest from the same start.

Use the earlier changed question as optional later practice. Reserve Debug a new ticket rule for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, **trace** conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Add 4 fits both reported results: from 4, 7 plus 4 minus 2 is 9; from 2, 5 plus 4 minus 2 is 7. Adding the original start predicts 5 on the second run and is inconsistent. Inspect the **actual** middle instruction.

If it says add 4, restore double the current score and expect 12 from start 4 and 8 from start 2; also check start 5 gives 14. This distinguishes the proposed faults without claiming no other cause exists. Exit: start 7.

Below 7 runs zero times and reports 7; at most 7 adds 2 once and reports 9.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the complete loop trace. Emphasise that the false test is a checkpoint that must be recorded.
7 to 20 minutes	Work the points-machine specification before revealing the observed run. Align checkpoints and find the first mismatch at the doubling step.
20 to 33 minutes	Compare add 4 with add original start. Predict outcomes from 2 before showing any observations. Explain the limits of choosing between two proposed causes.
33 to 45 minutes	Complete W1 and W2 in pairs. Ask why a passing result from start 1 does not remove the failure from start 3.
Session 2 start 0 to 8 minutes	Use W3 to distinguish repairing an instruction from replacing a reported answer. Retrieve the need for an unchanged specification.

Stage	Teacher action
8 to 23 minutes	Complete the guided repair and write expected retest results. Keep a record of the original failing case and prior passing cases.
23 to 38 minutes	Collect W4 independently. Check that starts are chosen before the add 1 step, so equality is reached from start 4.
38 to 45 minutes	Use W5 and the exit to practise a limited, evidence-based completion claim. Preview fitting a rule from examples next week.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

The maze's faulty trace reaches (0,1), then D attempts blocked (1,1) and fails before changing position. A valid repaired route is D,D,R,R: (0,0),(1,0),(2,0),(2,1),(2,2). That route also reaches the target when the blocked cell is removed; preserve both map versions as separate retests. Do not change the target to make a bad route appear correct. Reserve Debug a new ticket rule for the individual first attempt.

Run and record

1. Complete the task: Repair a maze algorithm and explain the exact failure.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

The last wrong number locates the fault: Compare every aligned checkpoint; an early mismatch can cause several later wrong values.

One revealing test proves the only possible cause: List the explanations **actual**ly compared. A different, untested fault may make the same **prediction**.

A correct answer on one **input** proves the repair: Use the constant-report example and rerun original failing and previously passing cases.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Debug a new ticket rule for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR Changing every part is the best way to find a cause.

For guessing, return to given facts and request a **trace**. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Expected results should follow **observed result**s: Calculate expectations from the **specification** first. Clarify genuine **specification** changes explicitly rather than hiding a failure.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A paper ticket rule must add 1 to the **input** count. If the updated count is at least 5, report Group; otherwise report Single. A copied version says above 5.

Choose the smallest three consecutive nonnegative starting counts that test just below, exactly at and just above the comparison boundary. Predict both versions, identify the revealing start and state the repair and retests.

Use starts 3, 4 and 5, which reach 4, 5 and 6. Intended results are Single, Group, Group. Copied results are Single, Single, Group. Start 4 exposes the equality fault.

Restore at least 5 and rerun all three, retaining starts 3 and 5 as previously passing regression cases.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible **trace**.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Locate the first difference

A procedure must add 2, double the updated value, then subtract 1. Starting at 3, an observed **trace** is 3, 5, 8, 7.

Write the **expected trace**, identify the first difference and explain why the final subtraction is not necessarily a separate fault.

Expected trace: 3, 5, 10, 9. First difference is the doubling checkpoint: 10 **expected** versus 8 observed. The later 8 minus 1 equals 7, so the observed subtraction is consistent with its **input**.

Investigate the earlier step before claiming another fault.

W2 Choose between two explanations

In W1 the faulty middle step may add 3, or add the original starting value. Both explain 5 becoming 8 when the start is 3. Choose start 1 or start 3 for a revealing new test.

Predict the final result under each explanation and under the intended procedure.

Choose start 1. After add 2 the value is 3. Adding 3 then subtracting 1 reports 5; adding original 1 then subtracting 1 reports 3. Doubling then subtracting 1 should report 5.

This separates the two fault explanations, but a final 5 alone cannot separate add 3 from correct doubling on this **input**; retain the original failing start 3.

W3 Reject a repair that hides the symptom

A learner repairs the original points machine by reporting 12 for every **input**, because 12 is correct from start 4. Give a different starting **input** and its **expected output** under add 3, double, subtract 2.

Explain why the constant report is not a valid repair and what earlier **evidence** should be preserved.

Start 2 should report 8, with **trace** 2, 5, 10, 8; other correctly calculated starts with a different **output** also work. Always reporting 12 breaks the **specification**.

Preserve the original **input** and observed intermediate **trace** as **evidence**, then correct the faulty instruction and rerun the cases.

W5 Make a defensible completion claim

After repairing W4, all three chosen tests pass. Write a short report with the fault, the exact repair, the three **input**s and **expected output**s, and what those passing tests establish.

Explain why saying every possible error is now impossible would go beyond the **evidence**.

The copied rule excluded equality. Change above 5 to at least 5 after the add 1 step. Tests 3, 4 and 5 now report Single, Group and Group.

The repair passes these checks around this boundary and retains the tested neighbouring behaviour. Other **input**s, missing values or unrelated implementation faults were not established by these three tests.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

It stops at square two because one move is missing. Add MOVE ONE before STOP and retest from the same start.

Additional transfer practice question

A sorting rule says blue goes left. A blue card goes right. The reader correctly reports blue, but the direction rule says blue goes right. Which part should you fix first?

Reasoned practice answer

Fix the direction rule, because the colour reading is already correct and the written mapping contradicts the intended behaviour. Retest the same blue card after the change.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Changing every part is the best way to find a cause.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Repeat three times, then stop. From start 0 the result must be 3; from start 2 it must be 5 if the requirement is to move three squares. Changing the target hides the bug.

Distinguish a fixed destination from a fixed travel distance before choosing the second test.

Explanations for the additional workbook examples

W1 Locate the first difference

W2 Choose between two explanations

W3 Reject a repair that hides the symptom

W4 Debug a new ticket rule

A paper ticket rule must add 1 to the **input** count. If the updated count is at least 5, report Group; otherwise report Single. A copied version says above 5.

Choose the smallest three consecutive nonnegative starting counts that test just below, exactly at and just above the comparison boundary. Predict both versions, identify the revealing start and state the repair and retests.

Use starts 3, 4 and 5, which reach 4, 5 and 6. Intended results are Single, Group, Group. Copied results are Single, Single, Group. Start 4 exposes the equality fault.

Restore at least 5 and rerun all three, retaining starts 3 and 5 as previously passing regression cases.

W5 Make a defensible completion claim

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

When the sorter fails, log the first wrong stage and change only the responsible rule, **input**, or label.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

When the sorter fails, write competing explanations and predicted **trace**s before changing code or instructions.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Provide aligned **expected** and observed columns and ask learners to point to the first unequal pair. Separate predicting a result from running the procedure.

Extension: explain why start 1 in W2 is useful for comparing the two faults but cannot alone prove correct doubling.

W4 is secure when the learner chooses starts 3, 4, 5, tests the updated values, explains equality, repairs the comparator and retains both neighbouring cases. W5 must limit the claim to the tested **evidence**.

After support, retry add 2 then if current is at most 6 report A otherwise B, with a below 6 fault. Starts 3, 4, 5 should produce A, A, B; the fault produces A, B, B.

Plan the delayed check

Suggested return points in this level: ai-1/13, ai-1/15, ai-1/18. Use **actual** classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 13: Learning from Examples. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 13 introduces **fitting** a pattern from examples. Students will carry forward the need to state candidate rules, compare their **prediction**s and keep **fitting evidence** distinct from later evaluation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case
</details>

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Compare **expected** and observed **trace**s to locate the first difference

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: In W1 the faulty middle step may add 3, or add the original starting value. Both explain 5 becoming 8 when the start is 3. Choose start 1 or start 3 for a revealing new test.

Predict the final result under each explanation and under the intended procedure.

Reasoned practice answer: Choose start 1. After add 2 the value is 3. Adding 3 then subtracting 1 reports 5; adding original 1 then subtracting 1 reports 3. Doubling then subtracting 1 should report 5.

This separates the two fault explanations, but a final 5 alone cannot separate add 3 from correct doubling on this **input**; retain the original failing start 3.

Check every learner before discussion: Use the worked case for Debugging with revealing tests. Proposed explanation: “The last wrong number locates the fault”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Compare every aligned checkpoint; an early mismatch can cause several later wrong values.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Use **input** 6 again instead of 2. Require a **prediction** and a reason before comparing. **Expected result**: Both hypotheses predict 12; the test remains inconclusive.

Reason: A revealing test requires different predicted **output**s, not merely another recorded run.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a **trace** of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, **trace**/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and **actual output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

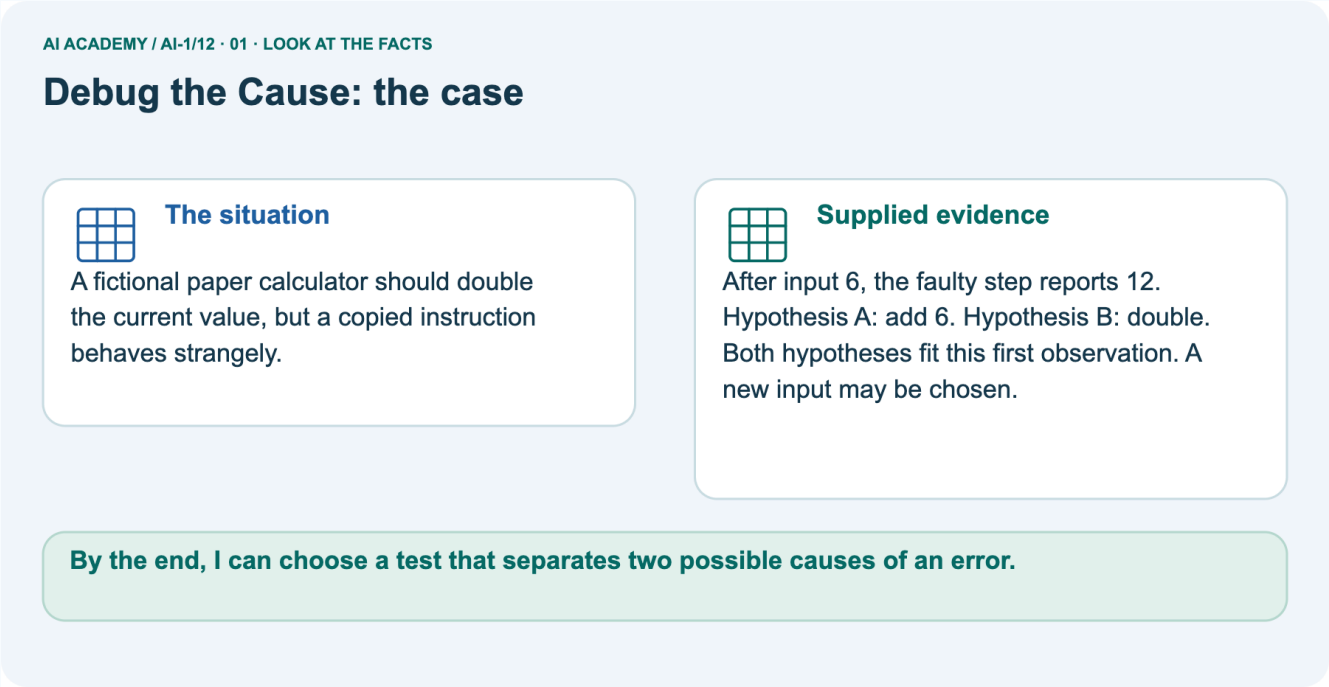
▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Debug the Cause: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

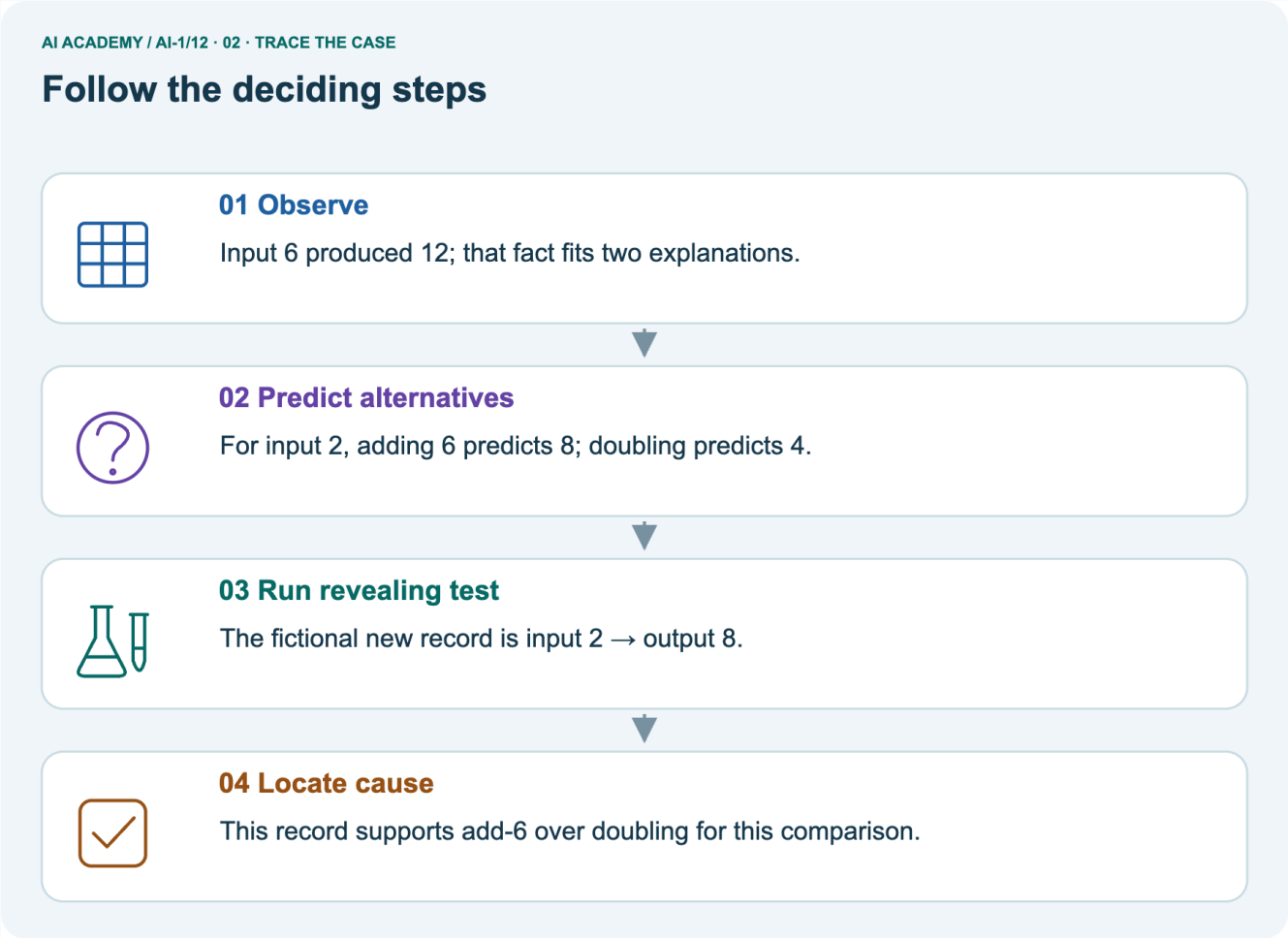
A fictional paper calculator should double the current value, but a copied instruction behaves strangely. Supplied facts: After **input** 6, the faulty step reports 12. Hypothesis A: add 6. Hypothesis B: double.

Both hypotheses fit this first **observation**. A new **input** may be chosen.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



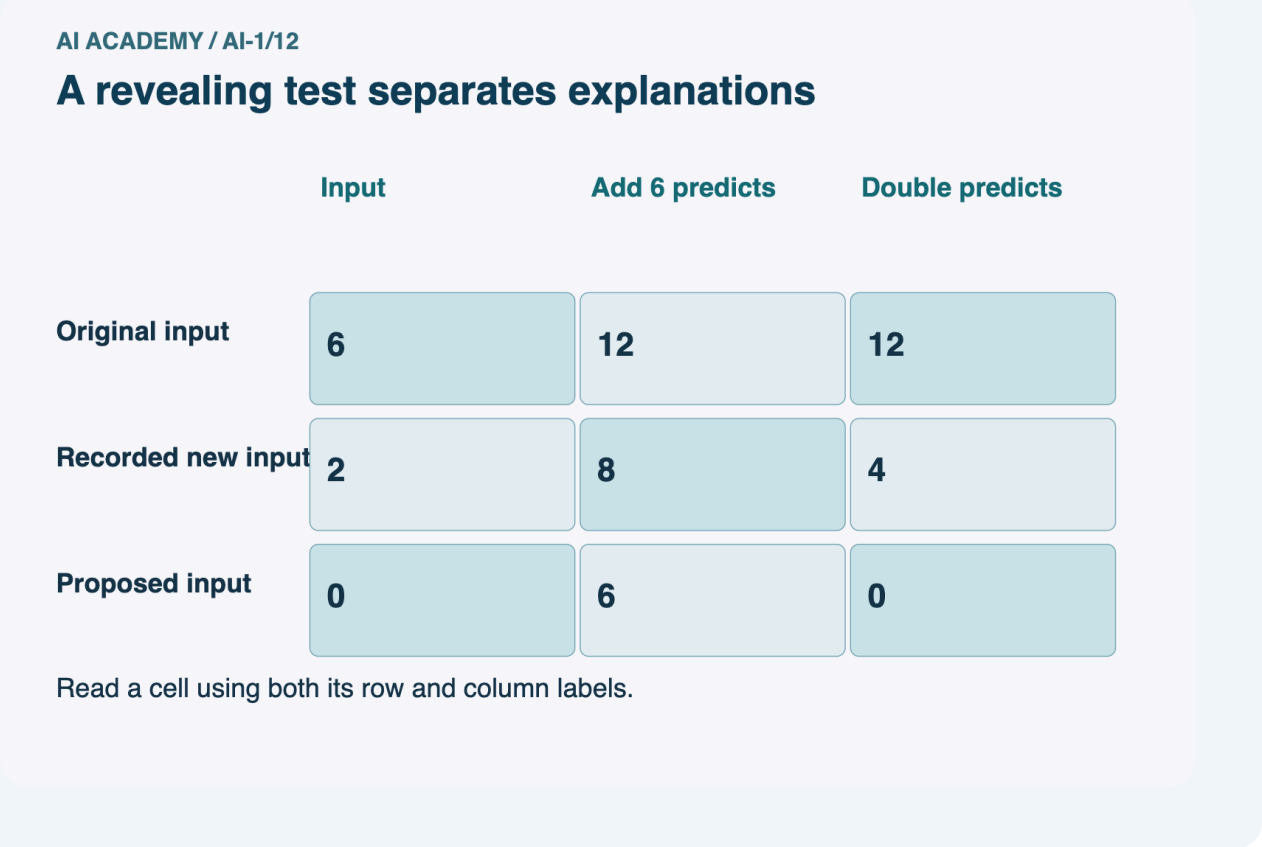
Follow the labelled picture with the matching explanation. Open the vector diagram

Observe: **Input** 6 produced 12; that fact fits two explanations. → Predict alternatives: For **input** 2, adding 6 predicts 8; doubling predicts 4. → Run revealing test: The fictional new record is **input** 2 → **output** 8. → Locate cause: This record supports add-6 over doubling for this comparison.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A revealing test separates explanations



Follow the labelled picture with the matching explanation. Open the vector diagram

A revealing test separates explanations. Original **input**: **Input** = 6, Add 6 predicts = 12, Double predicts = 12. Recorded new **input**: **Input** = 2, Add 6 predicts = 8, Double predicts = 4.

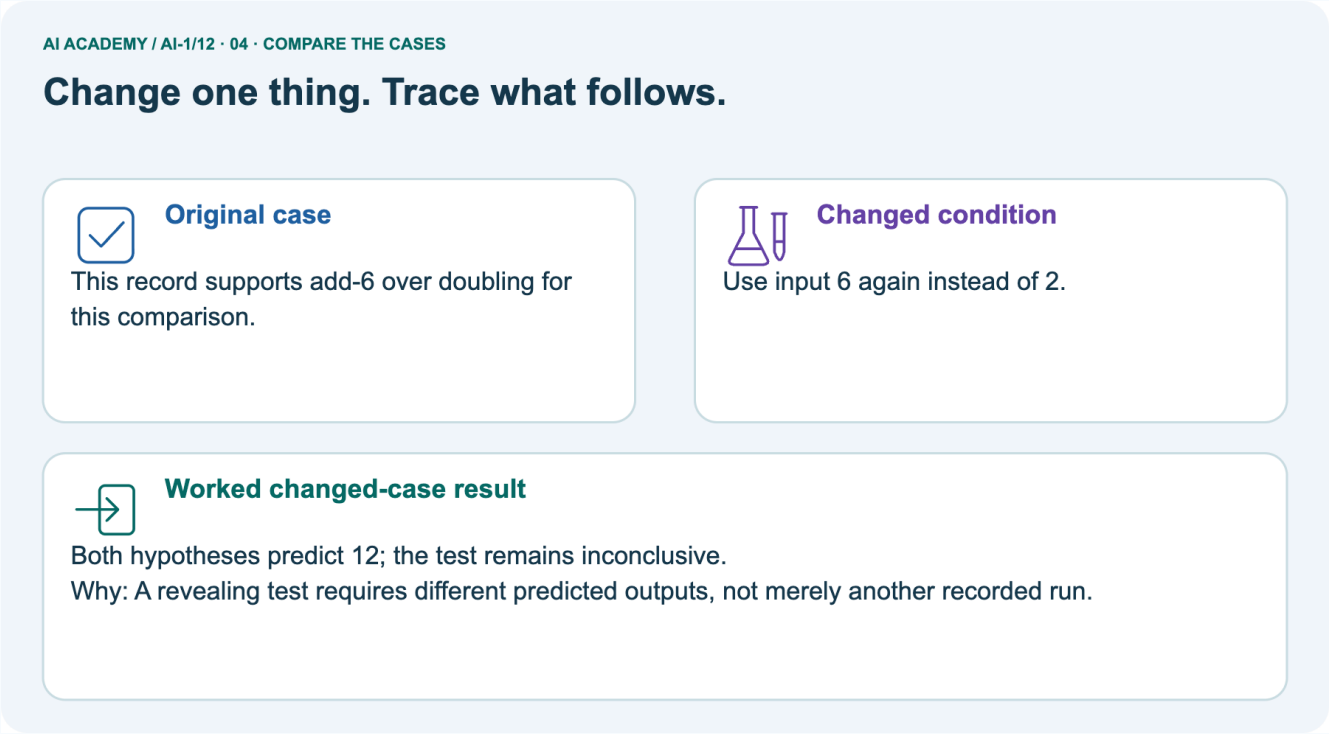
Proposed **input**: **Input** = 0, Add 6 predicts = 6, Double predicts = 0.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: This record supports add-6 over doubling for this comparison. Changed condition: Use **input** 6 again instead of 2. **Expected result**: Both hypotheses predict 12; the test remains inconclusive.

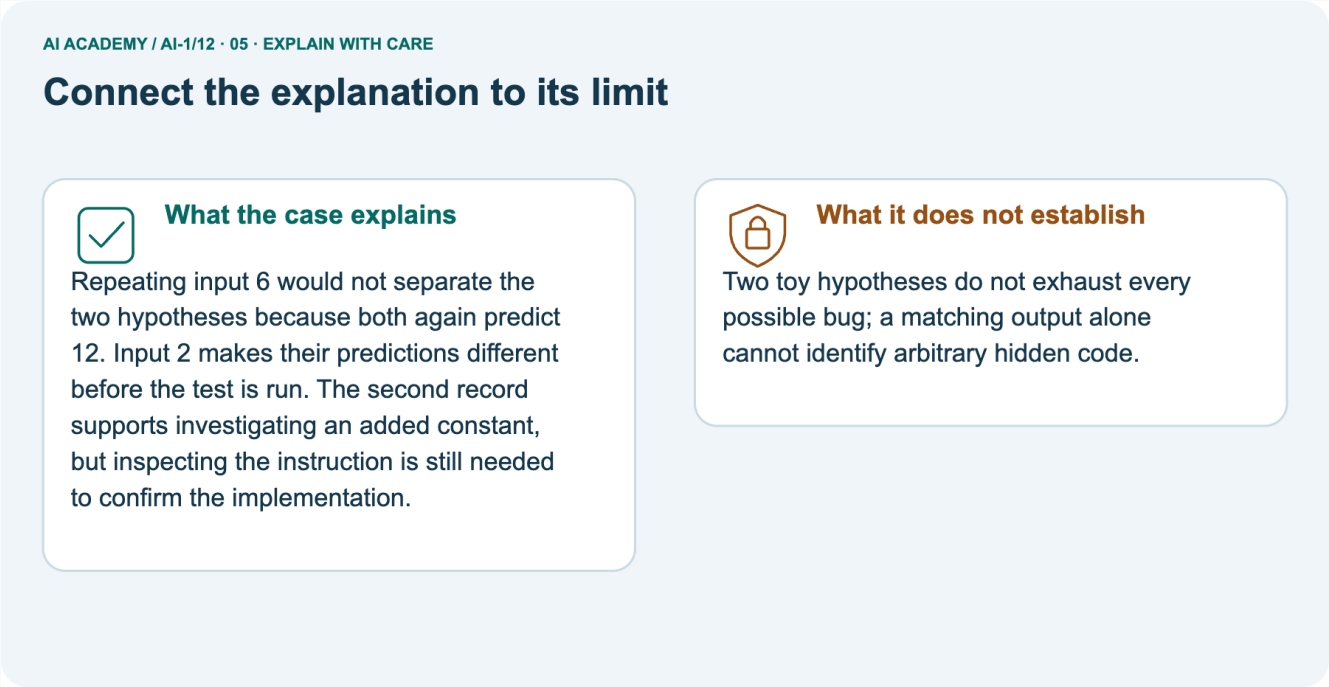
Reason: A revealing test requires different predicted **output**s, not merely another recorded run.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A revealing test requires different predicted **output**s, not merely another recorded run.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Repeating **input** 6 would not separate the two hypotheses because both again predict 12. **Input** 2 makes their **prediction**s different before the test is run.

The second record supports investigating an added constant, but inspecting the instruction is still needed to confirm the implementation.

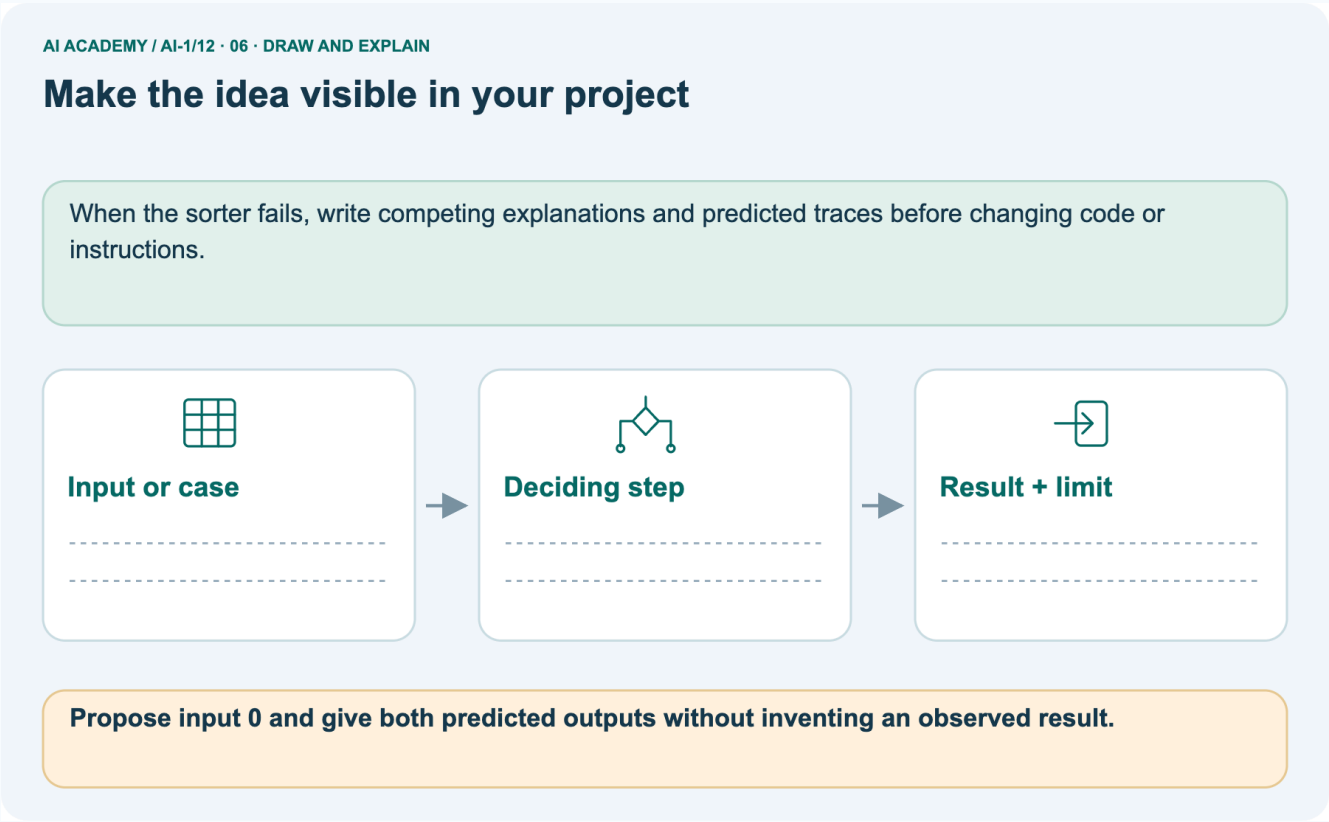
Limit: Two toy hypotheses do not exhaust every possible bug; a matching **output** alone cannot identify arbitrary hidden code.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Two toy hypotheses do not exhaust every possible bug; a matching **output** alone cannot identify arbitrary hidden code.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

When the sorter fails, write competing explanations and predicted **trace**s before changing code or instructions. Draw your **input**, deciding step, result and limit.

Supported question: Propose **input** 0 and give both predicted **output**s without inventing an **observed result**.

Draw your project **trace** and justify the deciding step using **evidence**.

Teacher prompt: Draw your project **trace** and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Add-6 predicts 6; doubling predicts 0. No **actual** result has been supplied for **input** 0.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Debug the Cause: the case

02. Follow the deciding steps

03. A revealing test separates explanations

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.